Domain and range

1

a

i Domain $= 8, 1, 3, 9, 2$

ii Range $= 5, 3, 4, 1, 9$

iii Function

b

i Domain $= 2, 3, 6, 8$

ii Range $= 3, 8, 1, 7, 2$

iii Not

c

i Domain $= 4, 8, 2, 7, 3$

ii Range $= 4, 8, 9, 1$

iii Function

d

i Domain $= 1, 6, 3, 8, 2$

ii Range $= 3, 2, 7, 1$

iii Function

e

i Domain $= 7, 8, 5, 3$

ii Range $= 1, 9, 3, 2, 6$

iii Not

f

i Domain $= 2, 4, 6, 8, 10$

ii Range$= 2, 4, 6, 8, 9$

iii Function

g

i Domain $= 3 \leq x \leq 7$

ii Range $= 5 \leq y \leq 8$

iii Function

h

i Domain $= 2 \leq x \leq 7$

ii Range $= 5$

iii Function

i

i Domain $= 5$

ii Range $= 2 \leq y \leq 6$

iii Not

j

i Domain $= 2 \leq x \leq 9$

ii Range $= 3 \leq y \leq 8$

iii Not

k

i Domain $= -7 \leq x \leq 7$

ii Range $= 0 \leq y \leq 7$

iii Function

l

i Domain $= -9 \leq x \leq 9$

ii Range $= -9 \leq y \leq 9$

iii Not

m

i All real x

ii $y \geq 0$

iii Function

n

i $x \geq 0$

ii All real y

iii Not

2

a

i Domain $= (-\infty, \infty)$

ii Range $= (-\infty, 2]$

b

i Domain $= (-\infty, \infty)$

ii Range $= [-2, \infty)$

c

i Domain $= [0, 6]$

ii Range $= [0, 7]$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

3

a False **b** False **c** True **d** True

4

a $(-\infty, \infty)$ **b** $(-\infty, \infty)$ **c** $[-6, 6]$

5

a 1 **b** No **c** All real x **d** $y > 0$

e $y = 128$ **f** $x = 8$

6

a 16 **b** $y \leq 16$ **c** All real x

7

a 0 **b** $y \geq 0$ **c** $x > 7$

8

a

i Minimum

ii Minimum value $= -1$

iii $y \geq -1$

b

i Maximum

ii Maximum value $= -3$

iii $y \leq -3$

9

a $x \geq -1$ **b** No

10

a Concave up **b** 5 **c** $y \geq 5$

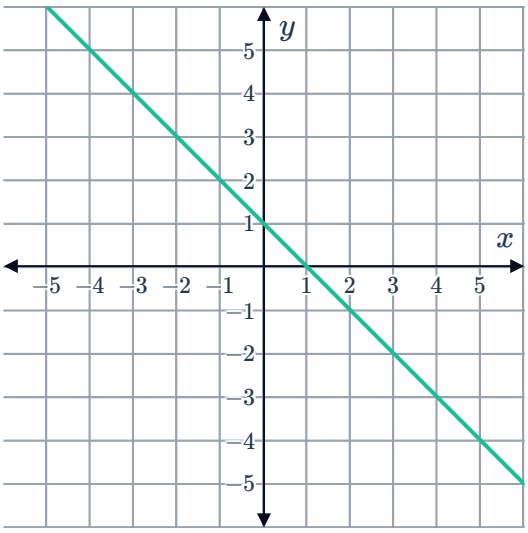
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Graphing functions

11

a

i

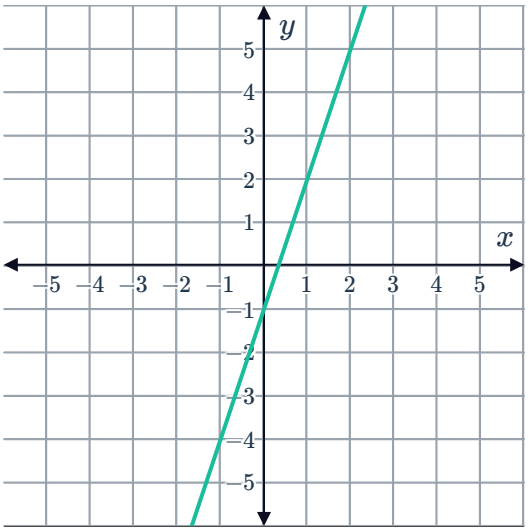


ii $(-\infty, \infty)$

iii $(-\infty, \infty)$

b

i



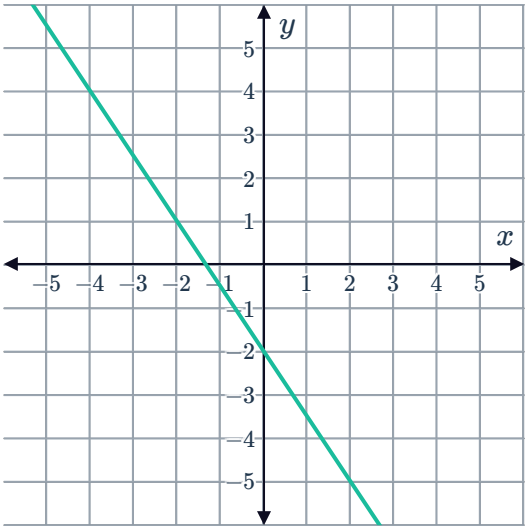
ii $(-\infty, \infty)$

iii $(-\infty, \infty)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c

i

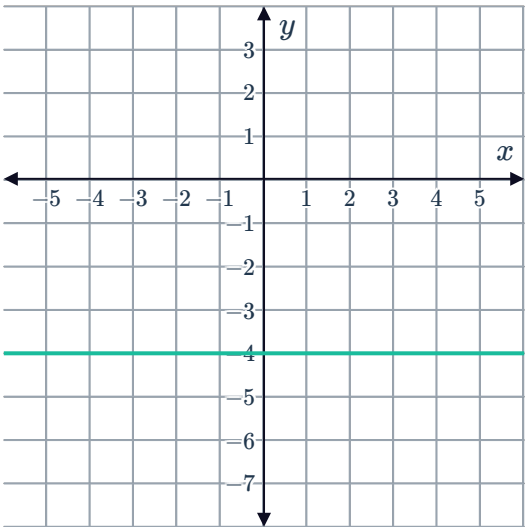


ii $(-\infty, \infty)$

iii $(-\infty, \infty)$

d

i



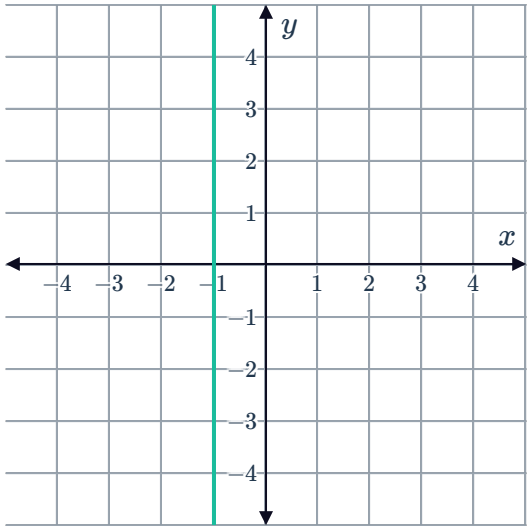
ii $(-\infty, \infty)$

iii -4

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

e

i



ii -1

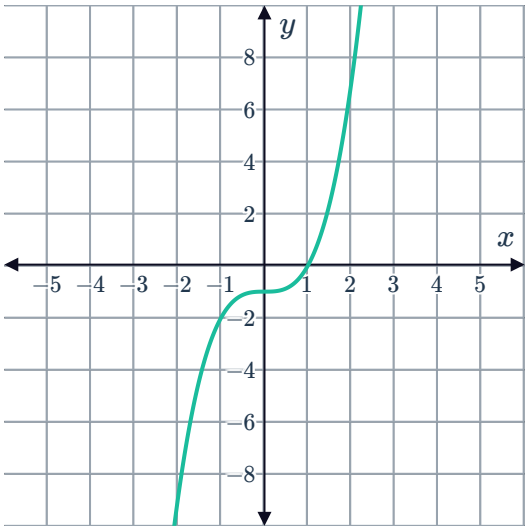
iii $(-\infty, \infty)$

12

a

x	-2	-1	0	1	2
y	-9	-2	-1	0	7

b



c Yes

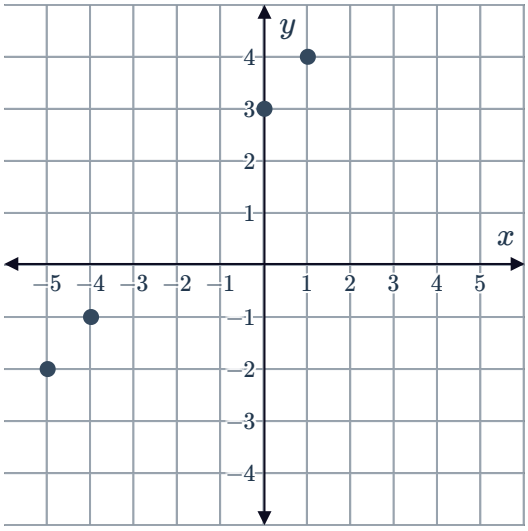
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13

a

x	-5	-4	0	1
$f(x)$	-2	-1	3	4

b

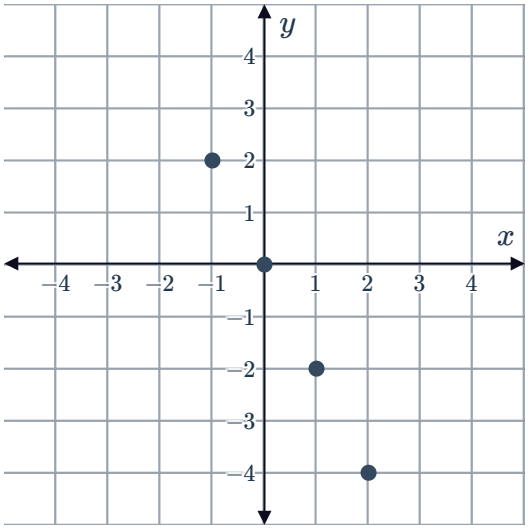


14

a

x	-1	0	1	2
$f(x)$	2	0	-2	-4

b

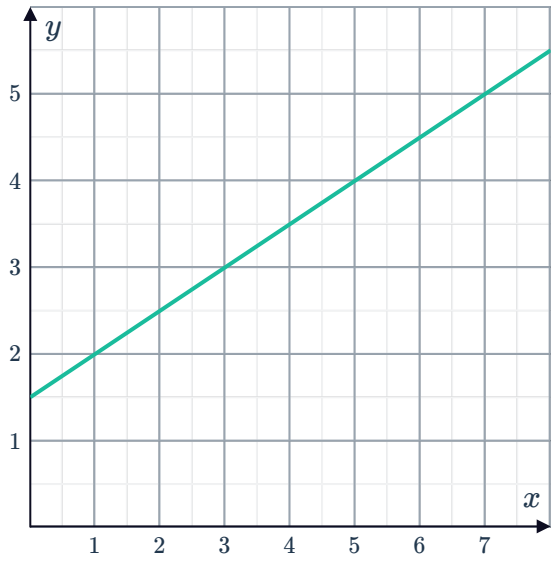


15

a

x	3	4	5	6	7
y	3	$\frac{7}{2}$	4	$\frac{9}{2}$	5

b



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

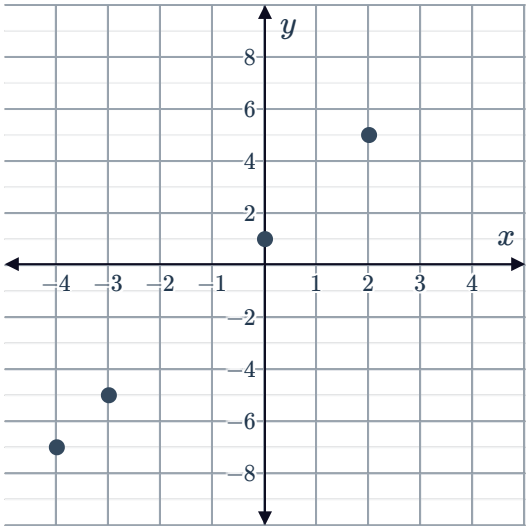
16

a $y = 2x + 1$

b

x	-4	-3	0	2
y	-7	-5	1	5

c

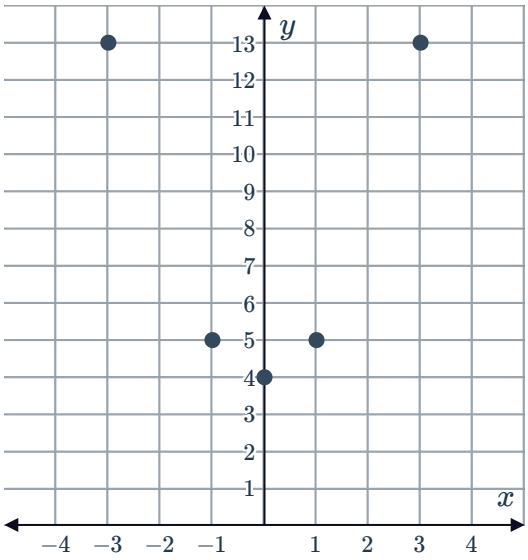


17

a

x	-3	-1	0	1	3
$f(x)$	13	5	4	5	13

b



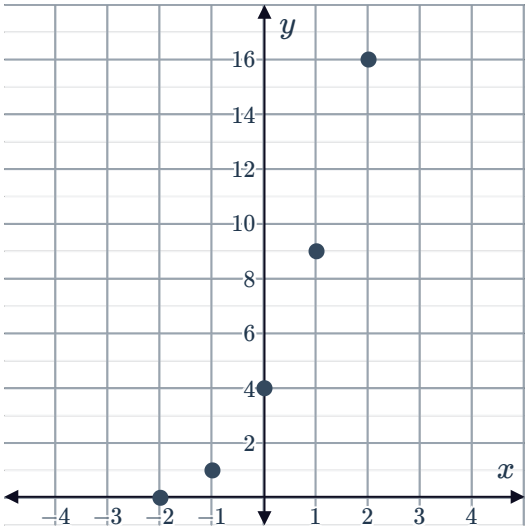
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

18

a

x	-2	-1	0	1	2
$f(x)$	0	1	4	9	16

b



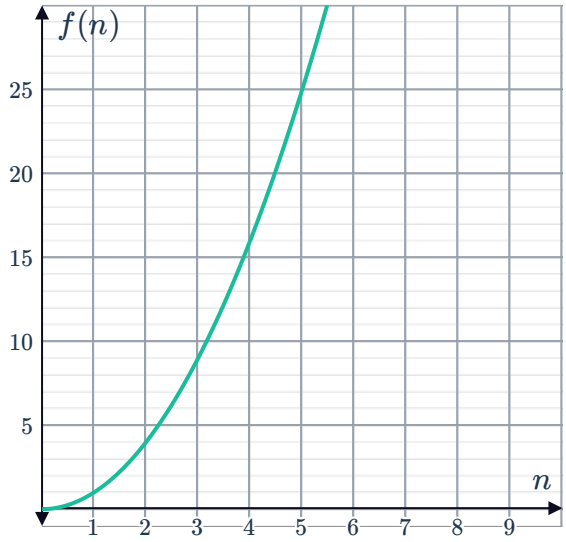
19

a

i No **ii** Yes **iii** Yes **iv** Yes

b $f(n) = n^2$

c



Rational functions

20 $(-\infty, -5) \cup (-5, \infty)$

21

a

i $(-\infty, 0) \cup (0, \infty)$ **ii** $(-\infty, 2) \cup (2, \infty)$

b

i $(-\infty, -6) \cup (-6, 6) \cup (6, \infty)$ **ii** $(-\infty, \infty)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Applications

22

a $0 \leq t \leq 5$

b $25 \leq T \leq 80$

c $0 \leq t \leq 3, 31 \leq t \leq 34$

d $0 \leq t \leq 3$

23

a $0 \leq x \leq 7$

b 30

c No

24

a $$f(x) = \begin{cases} -8, 0 \leq x < 1 \\ -15, 1 \leq x < 4 \\ -23, 4 \leq x \leq 10 \end{cases}$$

b $0 \leq x \leq 10$

c $-23^{\circ}C$

d 10 am

© ClipSAT: By Mr. Mohamed Abdallah +966597688647